05 / EXTRA MATHEMATICAL THEORY NOTE

EX29 — Why do we use the transpose $\mathbf{R}(\theta)^T$?

Part C: changing reference frames without rotating the force

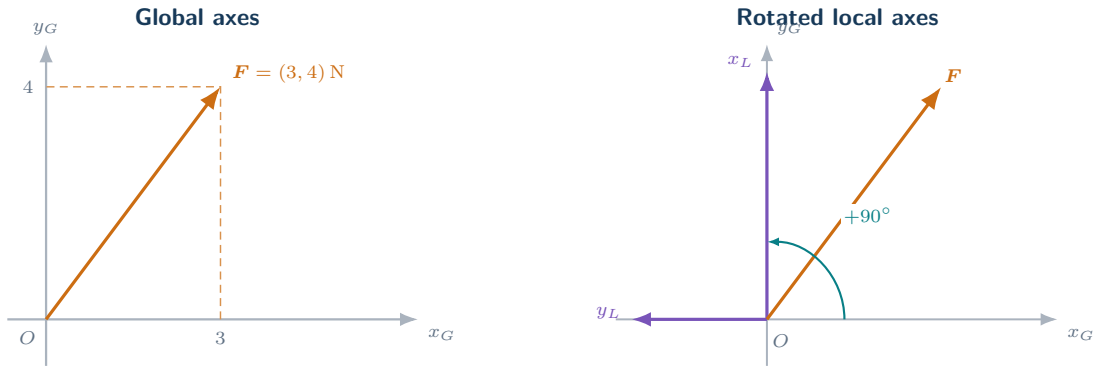
Core idea. In Part C, the force **does not rotate**. The **axes** used to describe its components rotate by $+90°$. The transpose converts global components into local components.

1. Two operations that may look alike, but are different

Active rotation: the coordinate axes stay fixed and the physical vector rotates. Rotating $\boldsymbol{v}$ counterclockwise by θ gives $\boldsymbol{v}' = \mathbf{R}(\theta)\boldsymbol{v}$.

Reference-frame change, or passive transformation: the physical vector stays fixed, but the local axes rotate by $+\theta$. The *numbers* describing the same vector change. No new physical force is created.

2. Geometric meaning: one force, two bases



The orange arrow represents exactly the same force in both drawings. Only the local axes are rotated in the right panel.

The global system uses axes (x_G, y_G). The local system (x_L, y_L) is obtained by rotating the global axes counterclockwise by $+\theta$. By convention, the **columns of $\mathbf{R}(\theta)$ are the local axes written in global coordinates**.

3. Constructing the matrix from the rotated axes

When the local axes rotate by $+\theta$, their unit vectors, written in global coordinates, are:

$$\boldsymbol{e}_{x,L} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad \boldsymbol{e}_{y,L} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$

Thus, putting these vectors in the **columns** gives:

$$\mathbf{R}(\theta) = \begin{bmatrix} \boldsymbol{e}_{x,L} & \boldsymbol{e}_{y,L} \end{bmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

A force with local components $(F_{L,x}, F_{L,y})$ can be reconstructed as a linear combination of the local axes:

$$\boldsymbol{F}_G = F_{L,x}\boldsymbol{e}_{x,L} + F_{L,y}\boldsymbol{e}_{y,L} = \mathbf{R}(\theta) \underbrace{\begin{pmatrix} F_{L,x} \\ F_{L,y} \end{pmatrix}}_{\boldsymbol{F}_L}.$$

This equation converts local to global: from known components along the local axes, it reconstructs the global components.

4. Why does the transpose appear? Step-by-step proof

In Part C, the *global* components are given and the *local* components are required. Invert the previous relation:

$$\boldsymbol{F}_G = \mathbf{R}(\theta)\boldsymbol{F}_L \quad \Longrightarrow \quad \boldsymbol{F}_L = \mathbf{R}(\theta)^{-1}\boldsymbol{F}_G.$$

A rotation matrix is **orthogonal**: its columns are unit vectors and mutually perpendicular. Writing $c = \cos\theta$ and $s = \sin\theta$ gives:

$$\mathbf{R}^T\mathbf{R} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} c^2+s^2 & -cs+sc \\ -sc+cs & s^2+c^2 \end{pmatrix} = \mathbf{I}.$$

Since $\cos^2\theta + \sin^2\theta = 1$, the transpose is exactly the inverse:

$$\boxed{\mathbf{R}(\theta)^{-1} = \mathbf{R}(\theta)^T = \mathbf{R}(-\theta)}$$

Substituting in the change-of-basis equation gives:

$$\boxed{\boldsymbol{F}_L = \mathbf{R}(\theta)^T\boldsymbol{F}_G}$$

Interpretation: the local axes were rotated by $+\theta$. To find the local coordinates of a fixed vector, apply the inverse transformation with angle $-\theta$.

5. Worked example from Part C: axes rotate by $+90^\circ$

The physical force remains fixed, with global components

$$\boldsymbol{F}_G = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \text{N}.$$

For $\theta = +90^\circ$, we have $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$:

$$\mathbf{R}(90^\circ) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{R}(90^\circ)^T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Applying the passive transformation gives:

$$\boldsymbol{F}_L = \mathbf{R}(90^\circ)^T \boldsymbol{F}_G = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \boxed{\begin{pmatrix} 4 \\ -3 \end{pmatrix} \text{N}}.$$

Where do 4 and −3 come from? The local axes are $\boldsymbol{e}_{x,L} = (0, 1)^T$ and $\boldsymbol{e}_{y,L} = (-1, 0)^T$. Each local component is the projection of the force onto its axis:

$$F_{L,x} = \boldsymbol{e}_{x,L} \cdot \boldsymbol{F}_G = 0(3) + 1(4) = 4, \quad F_{L,y} = \boldsymbol{e}_{y,L} \cdot \boldsymbol{F}_G = -1(3) + 0(4) = -3.$$

This offers another interpretation of $\mathbf{R}^T$: **its rows compute projections onto the local axes**.

6. Two checks to prevent common mistakes

1. **Recover the global components.** Transforming from local back to global:

$$\mathbf{R}(90^\circ)\boldsymbol{F}_L = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \text{N}.$$

2. **Preserve the magnitude.** It is the same physical force, so:

$$\|\boldsymbol{F}_G\|_2 = \sqrt{3^2 + 4^2} = 5 \text{ N}, \qquad \|\boldsymbol{F}_L\|_2 = \sqrt{4^2 + (-3)^2} = 5 \text{ N}.$$

What if we applied $\mathbf{R}(90^\circ)\boldsymbol{F}_G$ instead? The result would be $(-4, 3)^T$ N: this describes *physically rotating the force* by $+90^\circ$ while the global axes remain fixed, not changing the basis of the same fixed force.

Rule for EX29. If the **axes** rotate by $+\theta$ while the vector **does not rotate**, global-to-local conversion requires $\boxed{\boldsymbol{F}_L = \mathbf{R}(\theta)^T \boldsymbol{F}_G}$. To go back to global: $\boldsymbol{F}_G = \mathbf{R}(\theta)\boldsymbol{F}_L$. The identity $\mathbf{R}^{-1} = \mathbf{R}^T$ holds only because rotation matrices are orthogonal.

Looking ahead. This distinction will be essential when transforming positions, velocities, forces and sensor measurements between *body*, *navigation* and *inertial* frames in aerospace engineering and navigation.